

Kaustav Mukherjee

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EDUCATION

Carnegie Mellon University (CMU)

Master of Science in Computer Vision

Pittsburgh, PA

Aug. 2024 – Dec. 2025

National University of Singapore (NUS)

Bachelor of Engineering in Mechanical Engineering

Singapore, Singapore

Aug. 2020 – May 2024

EXPERIENCE

AI Engineer

Kaliber Labs

Jan 2026 – Present

San Francisco, CA

- Built a real-time 3D reconstruction pipeline from arthroscopic surgical video using TSDF fusion, directly resulting in a flagship product demo and a co-authored paper accepted at ECCV 2026 Workshop DCA-MI
- Improved textual and aural emotion recognition models by architecting a multi-task learning (MTL) framework for categorical classification and regression, driving a +30% absolute jump in accuracy
- Engineered a stateless microservice architecture with NATS and Redis for a multimodal conversational AI PoC
- Developed novel architectures for JEPA-style world models that improve baselines by over 50% on 3D benchmarks

AI Engineer Intern

Kaliber Labs

May 2025 – Aug. 2025

San Francisco, CA

- Architected a modular robotic navigation system with ROS2 and Nav 2, integrating 6+ CV models with C++ behavior trees via shared memory and multithreading, and tested on the LG Q9 robot and NVIDIA Isaac Sim
- Reduced per-frame edge processing latency from 300ms to 150ms across multiple CV inference tasks
- Led technical integration across 10+ team members and delivered 10+ live demos to investors and stakeholders

Software Engineering Intern

Kaliber Labs

Jan. 2023 – Jul. 2024

San Francisco, CA

- Fine-tuned stable diffusion to generate synthetic training data, expanding surgical tool datasets by up to 5 times
- Employed synthetic data generation and augmentation to train multiple image classifiers to above 97% accuracy
- Optimized model inference pipelines via ONNX and quantization, reducing memory footprint by 40%
- Fine-tuned Llama-3-8B on surgical data, achieving 84% of GPT-4o's performance for 2% of the operational cost
- Accelerated convergence of camera pose estimation algorithm by 10 times with particle swarm optimization (PSO)

Systems Engineering Intern

F-Drones

May 2022 – Oct. 2022

Singapore, Singapore

- Led team of 3 to overhaul drone communication system, increasing drone delivery consistency to over 95%
- Set up simulation-in-the-loop with Ardupilot, enabling diagnosis of a critical bug within 4 hours after a crash

PUBLICATIONS

Bychkov, D., Shah, R., Mukherjee, K., So, M. "Surface Coverage Completeness for Quality-Assured Arthroscopic 3D Reconstruction" *Proceedings of the ECCV Workshop on DCA-MI, 2026 (To appear)*.

PROJECTS

AvatarsFTW: 3D Human Avatars From the Wild | PyTorch, Image Generation

Oct 2024 – Dec 2024

- Created pipeline to improve performance of SOTA 3D human avatar generation models on in-the-wild images
- Developed novel method for human keypoint estimation to condition diffusion models for human image generation

Drone Perception System at CMU Airlab | Python, OpenCV, ROS, Computer Vision

Aug 2024 – Nov 2024

- Designed a perception system for a drone with a 2-DOF manipulator using ROS and RGBD cameras

SKILLS

Competencies: Generative AI (Diffusion, World Models), LLM Fine-tuning, Reinforcement Learning, 3D Computer Vision, Model Quantization, Context Engineering

Frameworks: PyTorch, Tensorflow, NumPy, OpenCV, ROS2

Tools: Docker, Redis, NATS, ONNX, AWS, Linux

Languages: Python, C++, JavaScript